FF-4DGS-Ego: hand pose, placement, and metric scale

Protocols differ. EgoGrasp, Dyn-HaMR and HaWoR evaluate over 128-frame world-space segments with SLAM at full resolution; ours is feedforward, per 16-frame clip, in the camera frame at 224 px. The numbers below are therefore not a direct head-to-head, and we do not claim to beat these methods on hand metrics.

Table 1. H2O hand pose, PA-MPJPE (mm, lower is better).

Method	PA-MPJPE	Protocol
Dyn-HaMR	**16.7**	world-space, SLAM (reported)
EgoGrasp	18.9	world-space, SLAM (reported)
HaWoR	30.8	world-space, SLAM (reported)
Ours (FF-4DGS-Ego)	46.8	feedforward, per-clip (clean subj1→4)
WiLoR (single-image)	47.6	our run

The Dyn-HaMR / EgoGrasp / HaWoR H2O PA-MPJPE values are from *EgoGrasp* (arXiv:2601.01050v2) Table 2 (PA-MPJPE column, 16.74/18.92/30.75; we round); the methods' own papers do not all self-report H2O PA-MPJPE, so cite EgoGrasp as the source.

Table 2. Our H2O numbers (21-joint, mm). PA on the clean subject1→4 split; the absolute columns are from the mixed eval set (clean-split re-eval pending).

	PA-MPJPE	C-MPJPE (abs.)	WRIST (abs.)
Ours (FF-4DGS-Ego)	46.8	58.0†	36.6†

† C-MPJPE / WRIST are from the mixed all-subjects eval set, *not* the held-out subject-4 split; a clean-split re-evaluation of the fine-tuned head is pending.

Table 3. Absolute placement gain from the 3D-joint loss (HOT3D, 9-seq split, mm).

	Absolute MPJPE	Hand depth
Shape-only loss (no abs-3D)	81	n/a
With absolute 3D-joint loss (ours)	**53** (−35%)	∼4.5 cm

Both rows report *absolute* MPJPE (no alignment); the Procrustes-aligned shape error (PA-MPJPE) is ≈ 8 mm and is not what this table measures.

Table 4. Metric scale and depth (lower is better).

Setting	Source	Depth error
Hand anchor (HOT3D)	in-scene MANO hand (ours)	**4.5 cm**
Hand anchor (HOT3D)	UniDepth-V2 (depth FM)	15.7 cm
GS scene depth (HOI4D)	no supervision	20.5 cm
GS scene depth (HOI4D)	with GT depth (ours, no mask)	**13.4 cm**
Object depth (HOT3D)	frozen, no metric loss	62 cm
Object depth (HOT3D)	frozen + hand anchor (sep. ckpt)	135 cm (degrades)

The two object-depth rows are *separate* frozen-backbone checkpoints (hand-anchor weight 0.0 vs 0.1); the anchor distorts non-hand-region depth when the backbone is frozen. The HOI4D 13.4 cm is best-so-far from an interrupted run (see Table 6 note).

Table 5. World-space placement, reported competitors (mm, lower is better).

Method	C-MPJPE	WA-MPJPE (short / long)	Dataset
HaWoR	51.8	22.5 / 27.4	HOI4D
Hand3R	42.6	38.0 / 56.7	HOI4D
Ours (FF-4DGS-Ego, SLAM-free)	93.0	**21.1** / 46.6	HOI4D

The HaWoR HOI4D row is a re-run reported in *Hand3R Table II* (arXiv:2602.03200), not HaWoR's own paper (which has no HOI4D / C-MPJPE); Hand3R's row is its own. C-MPJPE is absolute camera-frame (no alignment); WA-MPJPE aligns the trajectory (short/long = 30/100 frames). Ours: SLAM-free feedforward per-clip chaining (no DROID-SLAM), over 11 HOI4D sequences (ZY20210800001), one 128-frame segment each; absolute W-MPJPE = 353 mm, and WA-MPJPE (our per-window Umeyama Sim(3)) is 21/47 mm short/long, so the W-vs-WA gap is global chaining/scale drift. For reference, HaWoR reports HOT3D *W-MPJPE* 33.2 / WA 11.3 in its own Table 3 (a different dataset and metric, not directly comparable). Competitors use SLAM for the global trajectory.

Table 6. Scale-mechanism verdicts (trend over training, lower is better).

Route	Quantity	Trend
(b) feedforward scale head	metric-scale residual	14.1 → 13.1 cm
(a) partial unfreeze + GT depth	HOT3D object depth	23.8 → 22.3 cm
(a) HOI4D dense depth (best-so-far)	dense scene depth	**20.5 → 13.4 cm**

The (a) and (b) trends are directional, from short runs on a few sequences (storage-quota limited); the HOI4D row is best-so-far from a run cancelled at step 80 (residuals still oscillating 3–12 cm), *not* a converged number. They establish that the metric coupling is learnable (b) and that unfreezing pulls the scene toward metric rather than distorting it (a, opposite sign to the frozen-backbone baseline). Converged magnitudes are in progress.